Design and Evaluation of a Scalable Ticketing System using Microservices Architecture in High-Traffic Environment*

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Abstract

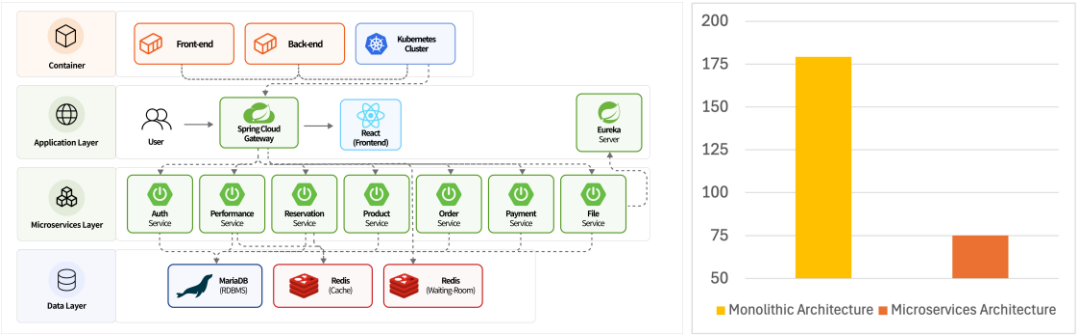
A ticketing system must be capable of handling a large volume of traffic generated within a short period of time. This study proposes a microservice-based ticketing system and a challenge-based macro bot detection method. The proposed approach enhances scalability by decomposing core functionalities into microservices. The performance evaluation results showed that the microservice-based system exhibited faster response times compared to the conventional monolithic service. This demonstrates that the system can respond more rapidly to large-scale requests.

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1 Introduction

An online ticketing system is a service in which a large number of users access simultaneously at the start of ticket sales. In particular, for popular concerts and sports events, seat inquiries, reservation requests, and payment requests are concentrated at a single point within a short period immediately after ticket opening. As a result, response delays, payment failures, and server outages may occur. Therefore, the ticketing system must operate reliably even under conditions of high request volumes. The conventional monolithic architecture simplifies development and deployment. However, when load is concentrated on a specific function, the overall service performance may degrade. For example, a surge in seat inquiries or reservation requests may also affect authentication and payment functionalities. In addition, it is difficult to scale only the required functionalities independently, making efficient resource utilization challenging [1]. To address these limitations, this study proposes a microservice-based ticketing system in which services are separated by functionality [2].

Figure 1: Architecture and Performance Evaluation of a Microservice-Based Ticketing System



2 Proposed Architecture

Figure 1 illustrates the proposed architecture and its performance. The architecture consists of authentication, event, reservation, payment, product, and order services. Each service can be deployed and scaled independently, and burst traffic is mitigated through queues under high traffic conditions. To evaluate the performance of the proposed system, load testing was conducted using k6 with 10,000 Virtual Users (VUs), and both the proposed and conventional approaches successfully processed the requests [3]. The system demonstrates a response time that is 2.39 times faster than that of the conventional monolithic architecture. This indicates that the microservice architecture can accommodate more users while maintaining equivalent processing outcomes. The system can respond promptly even when load is concentrated on specific functionalities, and it restricts automated access by applying challenge-based verification. The proposed system can improve response performance and service stability even under high traffic conditions.

This paper proposes a ticketing system based on a microservice architecture and demonstrates its superiority in terms of availability even under large-scale traffic. Future work will explore ways to extend the proposed architecture to handle 100,000 requests.

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